Prepare Dataset

Purpose

Prepare the raw VAERS data for downstream modeling and exploratory analysis by filtering to COVID-19 vaccine reports, aggregating records to the case level, engineering structured predictors, and exporting clean analysis-ready datasets.

Major sections

1. Setup and data loading
2. COVID-19 subsetting and case-level aggregation
3. Outcome construction and structured feature engineering
4. Exploratory data analysis
5. Rule-based comorbidity indicators

1. Setup and data loading

This section imports the analysis libraries, defines project-relative file paths, loads the raw VAERS source file, and inspects the starting data structure before any filtering or aggregation.

In [ ]:
```
# ============================================================
# 1.1 SETUP
# ============================================================

import os
import re
import sys
import json
import math
import warnings
from collections import defaultdict

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.dates as mdates

warnings.filterwarnings("ignore")

print("Python executable:", sys.executable)
print("Python version:", sys.version.split()[0])
print("Pandas version:", pd.__version__)
```

```
Python executable: /Users/ariellerothman/Desktop/Sperm Cell Analysis/.venv/b
in/python
Python version: 3.12.10
Pandas version: 2.2.3
```

In [ ]:

```python
# ============================================================
# 1.2 FILE PATHS
# ============================================================

from pathlib import Path

# Project root = the folder that contains this notebook.
# All output paths are resolved relative to it automatically.
HERE   = Path().resolve()
OUTDIR = str(HERE / "Outputs")
os.makedirs(OUTDIR, exist_ok=True)

# -- Raw VAERS data (NOT included in the repo - file is >1 GB) -------------
# Download from Kaggle: search 'VAERS ADE detection' or see README for the l
# Option A (recommended): set an environment variable before launching Jupyt
#   export VAERS_DATA_PATH=/absolute/path/to/COMBINED_DATA_ALL_YEARS.csv
# Option B: edit the fallback path below to match your local location.
DATA_PATH = os.environ.get(
    "VAERS_DATA_PATH",
    str(HERE.parent / "VAERS_data" / "COMBINED_DATA_ALL_YEARS.csv"),
)

print("DATA_PATH:", DATA_PATH)
print("OUTDIR   :", OUTDIR)
```

```
DATA_PATH: /Users/ariellerothman/Desktop/VAERS_data/COMBINED_DATA_ALL_YEARS.
csv
OUTDIR   : /Users/ariellerothman/Desktop/Capstone/Outputs
```

In [ ]:

```python
# ============================================================
# 1.3 LOAD RAW DATA
# ============================================================

df_raw = pd.read_csv(DATA_PATH, low_memory=False)

print("Raw shape:", df_raw.shape)
print(df_raw.head(3))
print("\nColumns:")
print(df_raw.columns.tolist())
```

```
Raw shape: (2132431, 55)
   VAERS_ID    RECVDATE STATE  AGE_YRS  CAGE_YR  CAGE_MO SEX RPT_DATE  \
0    810053  01/01/2020    OH     73.0     73.0      NaN   F      NaN
1    810053  01/01/2020    OH     73.0     73.0      NaN   F      NaN
2    855017  01/01/2020    HI     55.0     55.0      NaN   F      NaN

                                       SYMPTOM_TEXT DIED  ...  \
0  Extreme shivering for approx. 5 hours then fev...  NaN  ...
1  Extreme shivering for approx. 5 hours then fev...  NaN  ...
2  Symptoms occurred almost immediately- aching j...  NaN  ...

  SYMPTOMVERSION5 ORDER_y VAX_TYPE                    VAX_MANU  VAX_LOT  \
0            22.1       2     FLU3               SANOFI PASTEUR      NaN
1             NaN       2     FLU3               SANOFI PASTEUR      NaN
2             NaN       1   VARZOS  GLAXOSMITHKLINE BIOLOGICALS      NaN

  VAX_DOSE_SERIES VAX_ROUTE VAX_SITE  \
0             UNK       NaN      NaN
1             UNK       NaN      NaN
2               2       SYR       LA

                                 VAX_NAME ORDER
0  INFLUENZA (SEASONAL) (FLUZONE HIGH-DOSE)     2
1  INFLUENZA (SEASONAL) (FLUZONE HIGH-DOSE)     2
2                        ZOSTER (SHINGRIX)     1

[3 rows x 55 columns]

Columns:
['VAERS_ID', 'RECVDATE', 'STATE', 'AGE_YRS', 'CAGE_YR', 'CAGE_MO', 'SEX', 'R
PT_DATE', 'SYMPTOM_TEXT', 'DIED', 'DATEDIED', 'L_THREAT', 'ER_VISIT', 'HOSPI
TAL', 'HOSPDAYS', 'X_STAY', 'DISABLE', 'RECOVD', 'VAX_DATE', 'ONSET_DATE',
'NUMDAYS', 'LAB_DATA', 'V_ADMINBY', 'V_FUNDBY', 'OTHER_MEDS', 'CUR_ILL', 'HI
STORY', 'PRIOR_VAX', 'SPLTTYPE', 'FORM_VERS', 'TODAYS_DATE', 'BIRTH_DEFECT',
'OFC_VISIT', 'ER_ED_VISIT', 'ALLERGIES', 'ORDER_x', 'SYMPTOM1', 'SYMPTOMVERS
ION1', 'SYMPTOM2', 'SYMPTOMVERSION2', 'SYMPTOM3', 'SYMPTOMVERSION3', 'SYMPTO
M4', 'SYMPTOMVERSION4', 'SYMPTOM5', 'SYMPTOMVERSION5', 'ORDER_y', 'VAX_TYP
E', 'VAX_MANU', 'VAX_LOT', 'VAX_DOSE_SERIES', 'VAX_ROUTE', 'VAX_SITE', 'VAX_
NAME', 'ORDER']
```

In [ ]:

```python
# ============================================================
# 1.4 RAW DATA TYPES + MISSINGNESS SNAPSHOT
# ============================================================

print(df_raw.info())

missing_pct_raw = (df_raw.isna().mean() * 100).sort_values(ascending=False)
print("\nTop missingness percentages:")
print(missing_pct_raw.head(20))
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2132431 entries, 0 to 2132430
Data columns (total 55 columns):
 #   Column           Dtype
---  ------           -----
 0   VAERS_ID         int64
 1   RECVDATE         object
 2   STATE            object
 3   AGE_YRS          float64
 4   CAGE_YR          float64
 5   CAGE_MO          float64
 6   SEX              object
 7   RPT_DATE         object
 8   SYMPTOM_TEXT     object
 9   DIED             object
 10  DATEDIED         object
 11  L_THREAT         object
 12  ER_VISIT         object
 13  HOSPITAL         object
 14  HOSPDAYS         float64
 15  X_STAY           object
 16  DISABLE          object
 17  RECOVD           object
 18  VAX_DATE         object
 19  ONSET_DATE       object
 20  NUMDAYS          float64
 21  LAB_DATA         object
 22  V_ADMINBY        object
 23  V_FUNDBY         object
 24  OTHER_MEDS       object
 25  CUR_ILL          object
 26  HISTORY          object
 27  PRIOR_VAX        object
 28  SPLTTYPE         object
 29  FORM_VERS        int64
 30  TODAYS_DATE      object
 31  BIRTH_DEFECT     object
 32  OFC_VISIT        object
 33  ER_ED_VISIT      object
 34  ALLERGIES        object
 35  ORDER_x          int64
 36  SYMPTOM1         object
 37  SYMPTOMVERSION1  float64
 38  SYMPTOM2         object
 39  SYMPTOMVERSION2  float64
 40  SYMPTOM3         object
 41  SYMPTOMVERSION3  float64
 42  SYMPTOM4         object
 43  SYMPTOMVERSION4  float64
 44  SYMPTOM5         object
 45  SYMPTOMVERSION5  float64
 46  ORDER_y          int64
 47  VAX_TYPE         object
 48  VAX_MANU         object
 49  VAX_LOT          object
 50  VAX_DOSE_SERIES  object
```

```
 51  VAX_ROUTE        object
 52  VAX_SITE         object
 53  VAX_NAME         object
 54  ORDER            int64
dtypes: float64(10), int64(5), object(40)
memory usage: 894.8+ MB
None

Top missingness percentages:
ER_VISIT           99.957935
BIRTH_DEFECT       99.910149
X_STAY             99.908180
RPT_DATE           99.805621
V_FUNDBY           99.782127
CAGE_MO            97.539803
DATEDIED           97.280522
DIED               97.049893
L_THREAT           96.606690
DISABLE            96.125174
PRIOR_VAX          95.010530
HOSPDAYS           89.336865
HOSPITAL           84.349458
ER_ED_VISIT        81.581631
OFC_VISIT          75.273385
CUR_ILL            69.930188
SPLTTYPE           68.908115
SYMPTOM5           65.529717
SYMPTOMVERSION5    65.529717
LAB_DATA           58.832666
dtype: float64
```

2. COVID-19 subsetting and case-level aggregation

This section:

- subsets to `VAX_TYPE == "COVID19"`
- removes exact duplicate rows
- collapses multiple row-level records per `VAERS_ID`
- preserves clinically relevant information using conservative aggregation rules

In [ ]:

```
# ============================================================
# 2.1 SUBSET TO COVID-19 REPORTS
# ============================================================

df_covid_rows = df_raw.loc[df_raw["VAX_TYPE"] == "COVID19"].copy()

print("COVID row-level shape:", df_covid_rows.shape)
print("Unique VAERS_ID:", df_covid_rows["VAERS_ID"].nunique())
print("Extra rows beyond one-per-ID:", len(df_covid_rows) - df_covid_rows["V
```

```
COVID row-level shape: (1702653, 55)
Unique VAERS_ID: 986096
Extra rows beyond one-per-ID: 716557
```

In [ ]:
```
# ============================================================
# 2.2 REMOVE EXACT DUPLICATE ROWS
# ============================================================


before = len(df_covid_rows)
df_covid_rows = df_covid_rows.drop_duplicates()
after = len(df_covid_rows)

print("Rows before exact duplicate removal:", before)
print("Rows after exact duplicate removal:", after)
print("Exact duplicates removed:", before - after)
```

```
Rows before exact duplicate removal: 1702653
Rows after exact duplicate removal: 1699449
Exact duplicates removed: 3204
```

In [ ]:
```
# ============================================================
# 2.3 BASIC TYPE CLEANING BEFORE CASE-LEVEL AGGREGATION
# ============================================================

date_cols = ["RECVDATE", "VAX_DATE", "ONSET_DATE", "DATEDIED"]
for col in date_cols:
    if col in df_covid_rows.columns:
        df_covid_rows[col] = pd.to_datetime(df_covid_rows[col], errors="coer

# Binary outcome source columns in VAERS are usually Y/blank
binary_source_cols = [
    "DIED", "L_THREAT", "ER_VISIT", "ER_ED_VISIT",
    "HOSPITAL", "DISABLE", "BIRTH_DEFECT"
]

for col in binary_source_cols:
    if col in df_covid_rows.columns:
        df_covid_rows[col] = df_covid_rows[col].astype(str).str.upper().str.
        df_covid_rows[col] = df_covid_rows[col].replace({"Y": 1, "N": 0, "":
        df_covid_rows[col] = pd.to_numeric(df_covid_rows[col], errors="coerc

# Numeric columns
numeric_cols = ["AGE_YRS", "HOSPDAYS", "NUMDAYS"]
for col in numeric_cols:
    if col in df_covid_rows.columns:
        df_covid_rows[col] = pd.to_numeric(df_covid_rows[col], errors="coerc

# Standardize basic categorical text
for col in ["SEX", "STATE", "VAX_TYPE", "VAX_MANU", "VAX_NAME", "VAX_DOSE_SE
    if col in df_covid_rows.columns:
        df_covid_rows[col] = df_covid_rows[col].astype("string")
```

In [ ]:
```
# ============================================================
# 2.4 CASE-LEVEL AGGREGATION
# ============================================================
# Purpose: Collapse multiple rows per VAERS_ID into single case records

text_fields = ["SYMPTOM_TEXT", "LAB_DATA", "OTHER_MEDS", "CUR_ILL", "HISTORY
```

```
list_fields = ["VAX_MANU", "VAX_NAME", "VAX_DOSE_SERIES", "VAX_ROUTE", "VAX_

def concat_unique_text(series):
    """Concatenate unique text values from a series, preserving order."""
    vals = [str(x).strip() for x in series.dropna().astype(str) if str(x).st
    vals_unique = list(dict.fromkeys(vals))
    return " | ".join(vals_unique) if vals_unique else np.nan

def unique_list(series):
    """Extract unique non-null values from a series as a deduplicated list."
    vals = [str(x).strip() for x in series.dropna().astype(str) if str(x).st
    return list(dict.fromkeys(vals))

agg_dict = {}

for col in binary_source_cols:
    if col in df_covid_rows.columns:
        agg_dict[col] = "max"

if "RECVDATE" in df_covid_rows.columns:
    agg_dict["RECVDATE"] = "min"
if "VAX_DATE" in df_covid_rows.columns:
    agg_dict["VAX_DATE"] = "min"
if "ONSET_DATE" in df_covid_rows.columns:
    agg_dict["ONSET_DATE"] = "max"
if "DATEDIED" in df_covid_rows.columns:
    agg_dict["DATEDIED"] = "max"

if "AGE_YRS" in df_covid_rows.columns:
    agg_dict["AGE_YRS"] = "max"
if "HOSPDAYS" in df_covid_rows.columns:
    agg_dict["HOSPDAYS"] = "max"
if "NUMDAYS" in df_covid_rows.columns:
    agg_dict["NUMDAYS"] = "max"

for col in ["SEX", "STATE", "VAX_TYPE"]:
    if col in df_covid_rows.columns:
        agg_dict[col] = lambda s: s.dropna().iloc[0] if s.dropna().shape[0]

for col in text_fields:
    if col in df_covid_rows.columns:
        agg_dict[col] = concat_unique_text

for col in list_fields:
    if col in df_covid_rows.columns:
        agg_dict[col] = unique_list

df_case = df_covid_rows.groupby("VAERS_ID", as_index=False).agg(agg_dict)

print("Case-level shape:", df_case.shape)
print(df_case.head(2).T)
```

```
Case-level shape: (986096, 31)
                                                                  0  \
VAERS_ID                                                     109128
DIED                                                            NaN
L_THREAT                                                        NaN
ER_VISIT                                                        NaN
ER_ED_VISIT                                                     NaN
HOSPITAL                                                        1.0
DISABLE                                                         NaN
BIRTH_DEFECT                                                    NaN
RECVDATE                                          2021-03-18 00:00:00
VAX_DATE                                          2021-03-05 00:00:00
ONSET_DATE                                        2021-03-06 00:00:00
DATEDIED                                                        NaT
AGE_YRS                                                         NaN
HOSPDAYS                                                        NaN
NUMDAYS                                                         1.0
SEX                                                               M
STATE                                                            MS
VAX_TYPE                                                    COVID19
SYMPTOM_TEXT     Lungs filled with blood; A regulatory report f...
LAB_DATA                                                        NaN
OTHER_MEDS                                                      NaN
CUR_ILL                                                         NaN
HISTORY          Medical History/Concurrent Conditions: No adve...
PRIOR_VAX                                                       NaN
ALLERGIES                                                       NaN
VAX_MANU                                                  [MODERNA]
VAX_NAME                              [COVID19 (COVID19 (MODERNA))]
VAX_DOSE_SERIES                                                  [1]
VAX_ROUTE                                                       [OT]
VAX_SITE                                                        [LA]
VAX_LOT                                                    [026A21A]

                                                                  1
VAERS_ID                                                     151928
DIED                                                            NaN
L_THREAT                                                        NaN
ER_VISIT                                                        NaN
ER_ED_VISIT                                                     NaN
HOSPITAL                                                        NaN
DISABLE                                                         NaN
BIRTH_DEFECT                                                    NaN
RECVDATE                                          2021-03-31 00:00:00
VAX_DATE                                          2021-03-23 00:00:00
ONSET_DATE                                        2021-03-23 00:00:00
DATEDIED                                                        NaT
AGE_YRS                                                        56.0
HOSPDAYS                                                        NaN
NUMDAYS                                                         0.0
SEX                                                               F
STATE                                                            NJ
VAX_TYPE                                                    COVID19
SYMPTOM_TEXT     Approximately 10 min after receiving the injec...
LAB_DATA                                                        NaN
OTHER_MEDS       Multivitamin and vitamin D supplements | VITAM...
```

```
CUR_ILL                                Post nasal drip, but no illness
HISTORY          CAD;  h/o ovarian cancer | Medical History/Con...
PRIOR_VAX                                                        NaN
ALLERGIES                                                 Penicillin
VAX_MANU                                          [PFIZER\BIONTECH]
VAX_NAME                     [COVID19 (COVID19 (PFIZER-BIONTECH))]
VAX_DOSE_SERIES                                                  [1]
VAX_ROUTE                                                      [SYR]
VAX_SITE                                                        [RA]
VAX_LOT                                           [ER8727, Unknown]
```

3. Outcome construction and structured feature engineering

In [ ]:
```python
# ============================================================
# 3.1 OUTCOME CONSTRUCTION
# ============================================================

severity_components = [
    "DIED", "DISABLE", "L_THREAT", "HOSPITAL", "ER_VISIT", "ER_ED_VISIT", "E
]
severity_components = [c for c in severity_components if c in df_case.columr

for col in severity_components:
    df_case[col] = df_case[col].fillna(0).astype("int8")

df_case["SERIOUS"] = (df_case[severity_components].sum(axis=1) > 0).astype("
df_case["SEVERE"] = df_case["SERIOUS"].copy()

print(df_case[severity_components + ["SERIOUS"]].mean().sort_values(ascendir
```

```
SERIOUS         0.198506
ER_ED_VISIT     0.120642
HOSPITAL        0.089814
DISABLE         0.019798
DIED            0.019659
L_THREAT        0.016199
BIRTH_DEFECT    0.000647
ER_VISIT        0.000158
dtype: float64
```

In [ ]:
```python
# ============================================================
# 3.2 STRUCTURED FEATURE ENGINEERING
# ============================================================
# Purpose: Create derived features from raw fields for modeling
# Output: NEW columns like ONSET_INTERVAL, MAX_DOSE, MANU__*, etc.
# ============================================================

# Remove implausible ages
if "AGE_YRS" in df_case.columns:
    df_case.loc[(df_case["AGE_YRS"] < 0) | (df_case["AGE_YRS"] > 110), "AGE_

# Onset interval: Days between vaccination and symptom onset
# Clinically relevant: Short intervals suggest vaccine-related causality
if "ONSET_DATE" in df_case.columns and "VAX_DATE" in df_case.columns:
```

```python
    df_case["ONSET_INTERVAL"] = (df_case["ONSET_DATE"] - df_case["VAX_DATE"]

# Temporal features: Extract year and month from report receipt date
# Used for: Time-series analysis, controlling for temporal trends
if "RECVDATE" in df_case.columns:
    df_case["YEAR"] = df_case["RECVDATE"].dt.year.astype("Int64")
    df_case["MONTH"] = df_case["RECVDATE"].dt.month.astype("Int64")

# Dose features: Extract and standardize dose information
def dose_features(dose_list):
    """
    Extract structured dose metrics from a list of dose series values.

    Args:
        dose_list (list): Dose values (e.g., ["1", "2", "3"] or ["1", "2", "

    Returns:
        pd.Series with columns:
            - MAX_DOSE: Highest numeric dose (1, 2, 3, etc.).
            - DOSE_COUNT: Number of distinct dose types.
            - MULTI_DOSE: Binary flag (1 if 2+ distinct doses observed).
            - UNKNOWN_DOSE: Binary flag (1 if "UNK"/"UNKNOWN" in list).

    Logic:
        - "7+" and numeric values normalized to integers for comparison
        - Unknown doses flagged separately (indicates incomplete data)
        - Empty/null lists return [NA, 0, 0, 0]

    """
    if not isinstance(dose_list, list) or len(dose_list) == 0:
        return pd.Series([pd.NA, 0, 0, 0], index=["MAX_DOSE","DOSE_COUNT","M

    cleaned = []
    unknown = 0
    for d in dose_list:
        if pd.isna(d):
            continue
        d = str(d).strip().upper()
        if d in ("UNK", "UNKNOWN"):
            unknown = 1
            cleaned.append("UNK")
        elif d == "7+":
            cleaned.append("7+")
        else:
            cleaned.append(d)

    # Convert cleaned values to numeric for aggregation
    nums = []
    for d in cleaned:
        if d == "7+":
            nums.append(7)
        else:
            try:
                nums.append(int(d))
            except Exception:
                pass
```

```
    max_dose = max(nums) if nums else pd.NA
    dose_count = len(set(cleaned))
    multi_dose = int(dose_count > 1)

    return pd.Series([max_dose, dose_count, multi_dose, unknown], index=["MA

if "VAX_DOSE_SERIES" in df_case.columns:
    df_case[["MAX_DOSE","DOSE_COUNT","MULTI_DOSE","UNKNOWN_DOSE"]] = df_case

# Manufacturer multi-hot encoding
# Purpose: Create binary indicators for each vaccine manufacturer observed i
# Rationale: Multi-manufacturer exposure (booster switching) is clinically m
if "VAX_MANU" in df_case.columns:
    manu_lists = df_case["VAX_MANU"].apply(lambda x: x if isinstance(x, list
    manufacturers = sorted(set(m for lst in manu_lists for m in lst if pd.no

    # Create one binary column per manufacturer
    # E.g., MANU__PFIZER=1 if Pfizer was administered, 0 otherwise
    for manu in manufacturers:
        safe = re.sub(r"[^A-Za-z0-9]+", "_", str(manu)).strip("_").upper()
        colname = f"MANU__{safe}"
        df_case[colname] = manu_lists.apply(lambda lst: int(manu in set(lst)

    # Flag cases with multiple manufacturers (e.g., dose 1=Pfizer, dose 2=Mo
    df_case["MULTI_MANUFACTURER"] = manu_lists.apply(lambda lst: int(len(set

print("Manufacturer columns:", [c for c in df_case.columns if c.startswith("
print(df_case[["MAX_DOSE","DOSE_COUNT","MULTI_DOSE","UNKNOWN_DOSE","MULTI_MA
```

```
Manufacturer columns: ['MANU__JANSSEN', 'MANU__MODERNA', 'MANU__NOVAVAX', 'M
ANU__PFIZER_BIONTECH', 'MANU__UNKNOWN_MANUFACTURER']
  MAX_DOSE  DOSE_COUNT  MULTI_DOSE  UNKNOWN_DOSE  MULTI_MANUFACTURER
0        1           1           0             0                   0
1        1           1           0             0                   0
2        1           1           0             0                   0
3        1           1           0             0                   0
4        1           1           0             0                   0
```

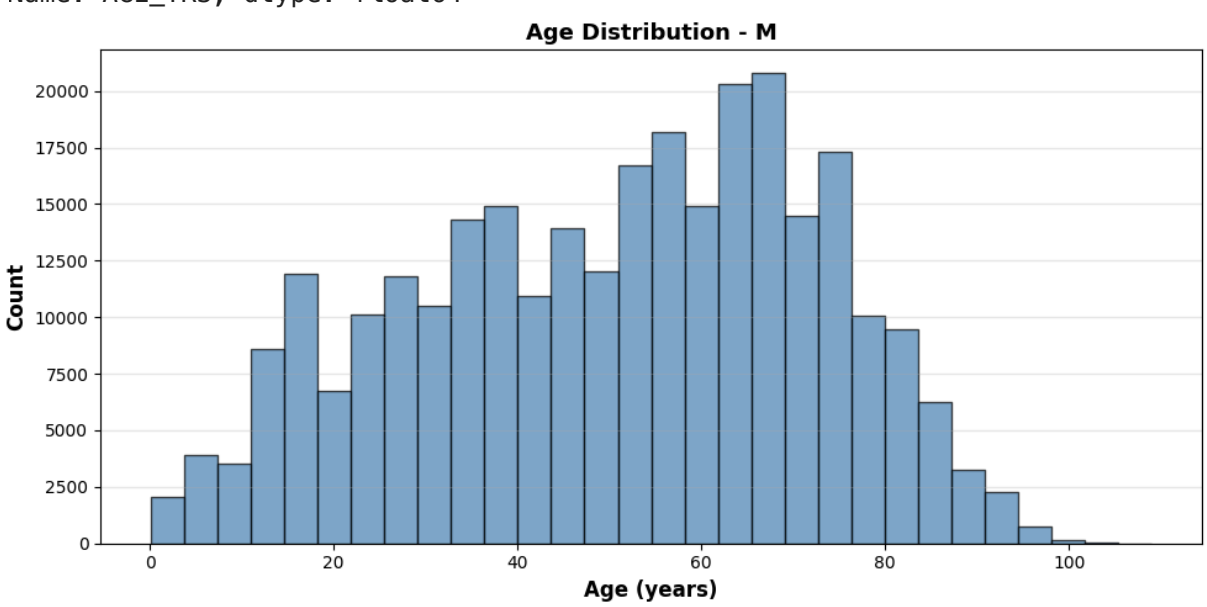
4. Exploratory data analysis

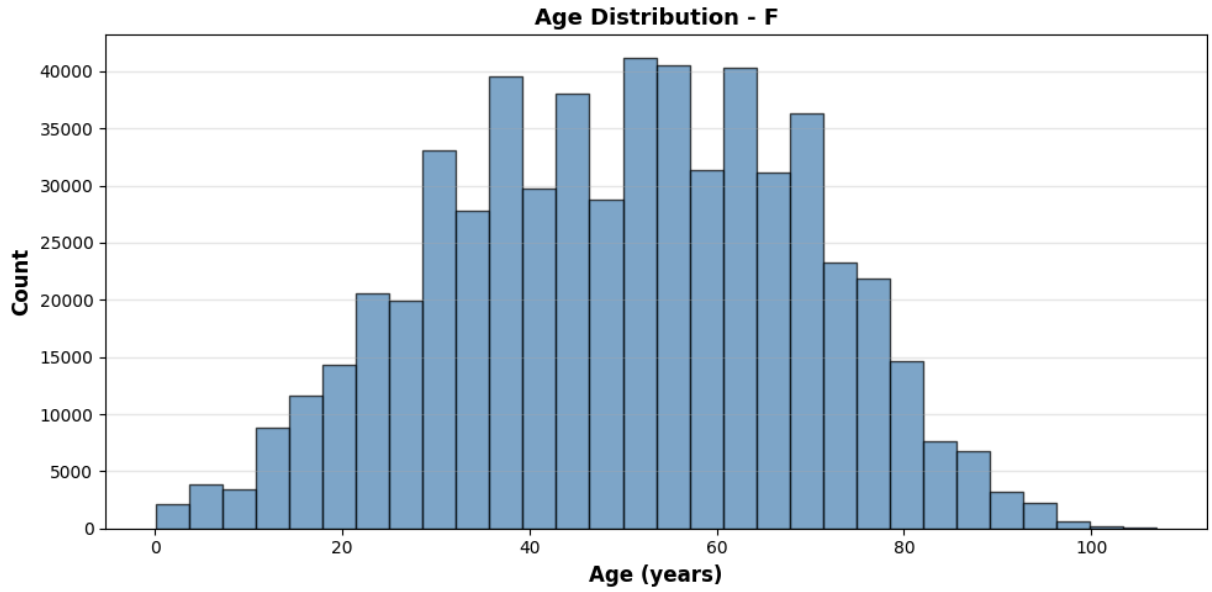
In [ ]:

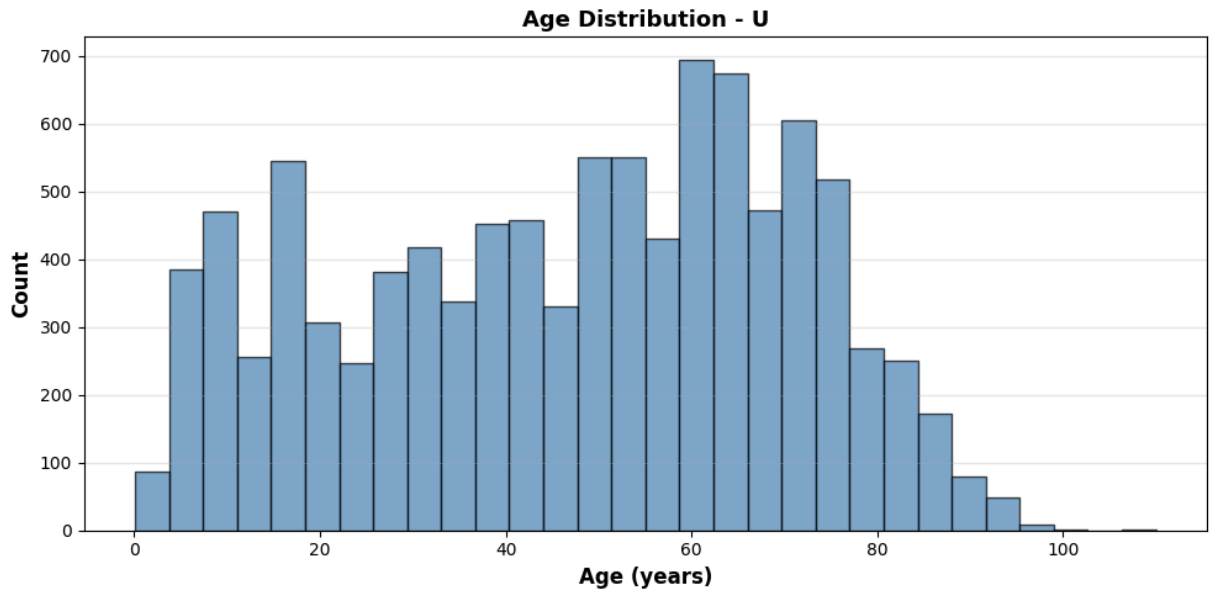
```
# ============================================================
# 4.1 BASIC DESCRIPTIVE STATISTICS
# ============================================================

print("Case-level dataset shape:", df_clean.shape)
print("\nSerious rate:", df_clean["SERIOUS"].mean())

if "SEX" in df_clean.columns:
    print("\nSex distribution:")
    print(df_clean["SEX"].value_counts(dropna=False))

if "AGE_YRS" in df_clean.columns:
    print("\nAge summary:")
    print(df_clean["AGE_YRS"].describe())
```

```
    if "SEX" in df_clean.columns:
        for sex in df_clean["SEX"].unique():
            if pd.notna(sex):
                data = df_clean[df_clean["SEX"] == sex]["AGE_YRS"].dropna()

                plt.figure(figsize=(10, 5))
                plt.hist(data, bins=30, color="steelblue", edgecolor="black"
                plt.xlabel("Age (years)", fontsize=12, fontweight="bold")
                plt.ylabel("Count", fontsize=12, fontweight="bold")
                plt.title(f"Age Distribution - {sex}", fontsize=13, fontweig
                plt.grid(axis='y', alpha=0.3)
                plt.tight_layout()
                plt.show()
```

```
Case-level dataset shape: (986096, 51)

Serious rate: 0.1985060278106797

Sex distribution:
SEX
F       630052
M       314301
U        41509
<NA>       234
Name: count, dtype: Int64

Age summary:
count    883244.000000
mean         50.006935
std          20.166280
min           0.080000
25%          35.000000
50%          51.000000
75%          66.000000
max         110.000000
Name: AGE_YRS, dtype: float64
```



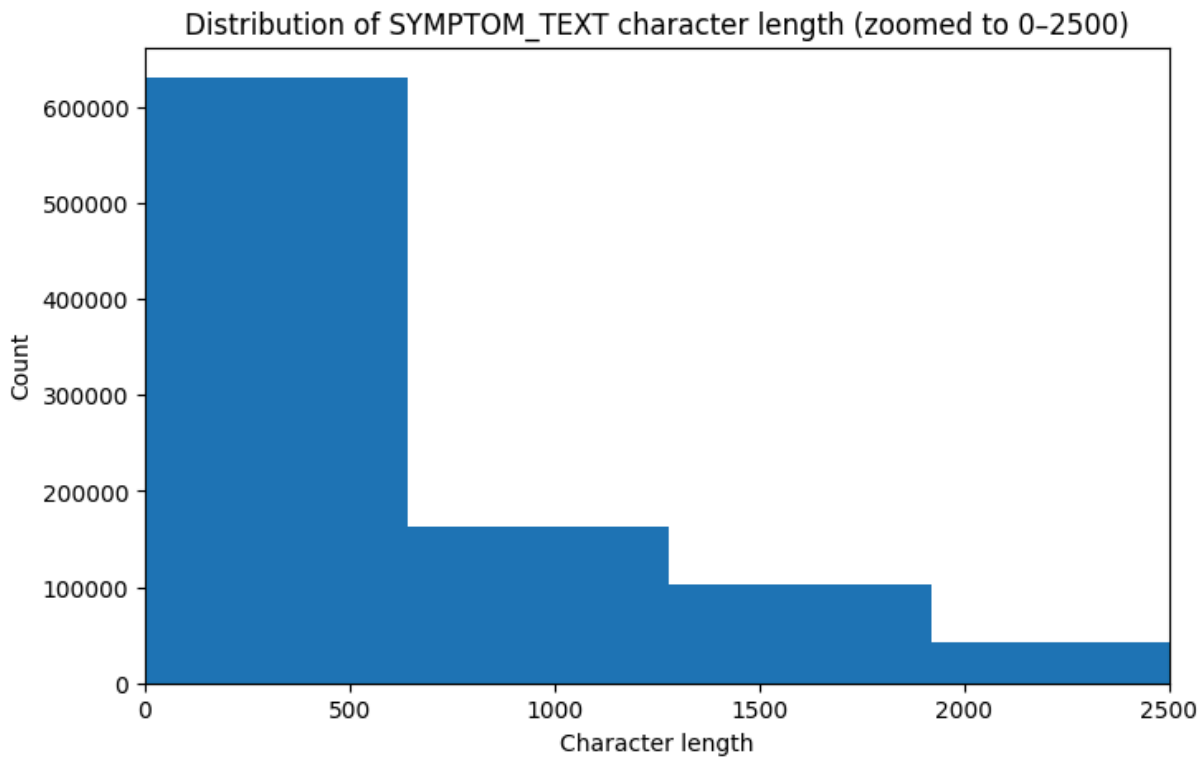




In [ ]:

```
# ============================================================
# 4.2 SYMPTOM NARRATIVE LENGTH
# ============================================================

if "SYMPTOM_TEXT" in df_clean.columns:
    df_clean["SYMPTOM_TEXT_CHAR_LEN"] = df_clean["SYMPTOM_TEXT"].fillna("").
    print(df_clean["SYMPTOM_TEXT_CHAR_LEN"].describe())

    plt.figure(figsize=(8,5))
    plt.hist(df_clean["SYMPTOM_TEXT_CHAR_LEN"], bins=100)
    plt.xlim(0, 2500)
    plt.title("Distribution of SYMPTOM_TEXT character length (zoomed to 0-25
    plt.xlabel("Character length")
    plt.ylabel("Count")
    plt.show()
```

```
count    986096.000000
mean        726.199905
std        1038.736049
min           0.000000
25%         101.000000
50%         312.000000
75%        1051.000000
max       63998.000000
Name: SYMPTOM_TEXT_CHAR_LEN, dtype: float64
```



In [ ]:

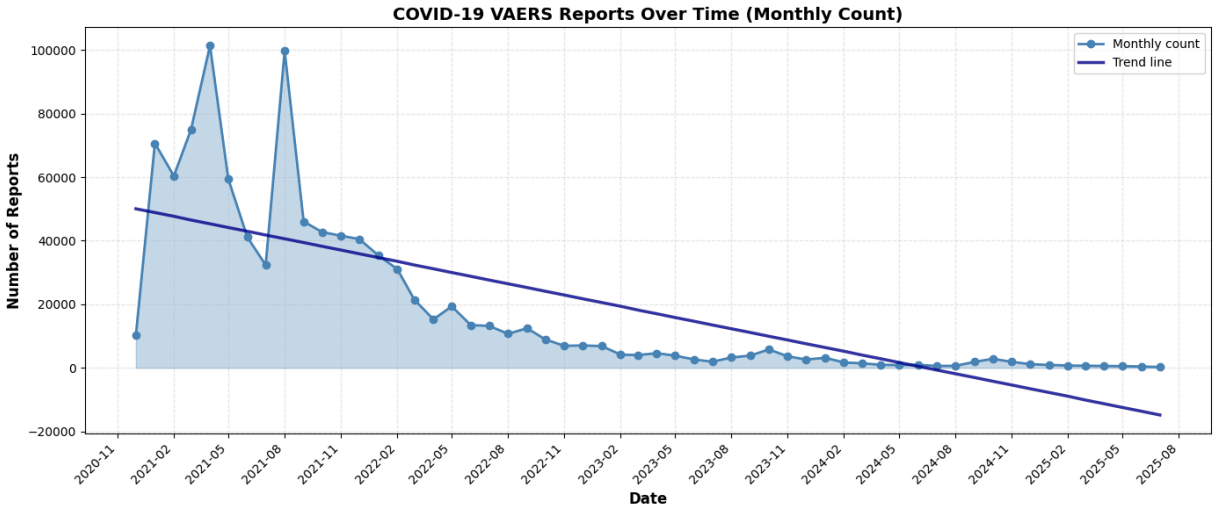
```
# ============================================================
# 4.3 TIME SERIES ANALYSIS: MONTHLY COVID REPORTS & OUTCOMES
# ============================================================

print("\n" + "="*80)
print("TIME SERIES ANALYSIS: MONTHLY COVID REPORTS & SERIOUS OUTCOMES")
print("="*80)

if "RECVDATE" in df_clean.columns:
    df_clean["RECVDATE"] = pd.to_datetime(df_clean["RECVDATE"], errors="coer

if "YEAR" not in df_clean.columns:
    df_clean["YEAR"] = df_clean["RECVDATE"].dt.year
if "MONTH" not in df_clean.columns:
    df_clean["MONTH"] = df_clean["RECVDATE"].dt.month

df_clean["YEAR_MONTH"] = df_clean["RECVDATE"].dt.to_period("M")

print("\n4.3A. MONTHLY NUMBER OF COVID-19 REPORTS")
print("-" * 80)

monthly_counts = df_clean.groupby("YEAR_MONTH").size().reset_index(name="rep
monthly_counts["YEAR_MONTH"] = monthly_counts["YEAR_MONTH"].dt.to_timestamp(
```

```
print(f"\nDate range: {monthly_counts['YEAR_MONTH'].min().date()} to {monthl
print(f"Total months: {len(monthly_counts)}")
print(f"Total reports: {monthly_counts['report_count'].sum():,}")
print(f"\nMonthly statistics:")
print(f"  Mean:     {monthly_counts['report_count'].mean():.0f} reports/mont
print(f"  Median:   {monthly_counts['report_count'].median():.0f} reports/mo
print(f"  Min:      {monthly_counts['report_count'].min():,} (in {monthly_co
print(f"  Max:      {monthly_counts['report_count'].max():,} (in {monthly_co

fig, ax = plt.subplots(figsize=(14, 6))

ax.plot(monthly_counts["YEAR_MONTH"], monthly_counts["report_count"],
        marker="o", linewidth=2, markersize=6, color="steelblue", label="Mor
ax.fill_between(monthly_counts["YEAR_MONTH"], monthly_counts["report_count"]
                alpha=0.3, color="steelblue")

x_numeric = np.arange(len(monthly_counts))
z = np.polyfit(x_numeric, monthly_counts["report_count"], 1)
p = np.poly1d(z)
ax.plot(monthly_counts["YEAR_MONTH"], p(x_numeric), "darkblue", linewidth=2.
        label="Trend line", alpha=0.8)

ax.set_xlabel("Date", fontsize=12, fontweight="bold")
ax.set_ylabel("Number of Reports", fontsize=12, fontweight="bold")
ax.set_title("COVID-19 VAERS Reports Over Time (Monthly Count)", fontsize=14
ax.grid(True, alpha=0.3, linestyle="--")
ax.legend(fontsize=10)

ax.xaxis.set_major_locator(mdates.MonthLocator(interval=3))
ax.xaxis.set_major_formatter(mdates.DateFormatter("%Y-%m"))
plt.xticks(rotation=45, ha="right")

plt.tight_layout()
plt.savefig(f"{OUTDIR}/01_monthly_report_count.png", dpi=300, bbox_inches="t
print(f"\nSaved: {OUTDIR}/01_monthly_report_count.png")
plt.show()

monthly_counts.to_csv(f"{OUTDIR}/monthly_report_count.csv", index=False)
print(f"Saved: {OUTDIR}/monthly_report_count.csv")

# MONTHLY SERIOUS OUTCOME PROPORTION
print("\n\n4.3B. MONTHLY PROPORTION OF SERIOUS REPORTS")
print("-" * 80)

if "SERIOUS" in df_clean.columns:
    monthly_serious = df_clean.groupby("YEAR_MONTH").agg(
        serious_count=("SERIOUS", "sum"),
        total=("SERIOUS", "count")
    ).reset_index()
    monthly_serious["serious_percent"] = monthly_serious["serious_count"] /
    monthly_serious["YEAR_MONTH"] = monthly_serious["YEAR_MONTH"].dt.to_time

    fig, ax = plt.subplots(figsize=(14, 6))

    ax.plot(monthly_serious["YEAR_MONTH"], monthly_serious["serious_percent"
```

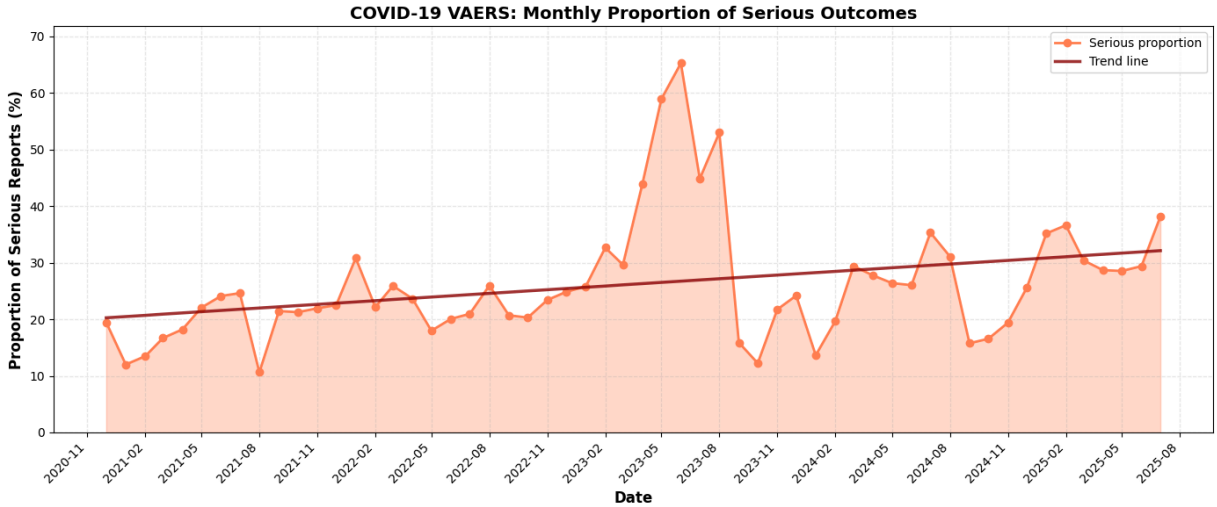
```
            marker="o", linewidth=2, markersize=6, color="coral", label="Ser
    ax.fill_between(monthly_serious["YEAR_MONTH"], monthly_serious["serious_
                    alpha=0.3, color="coral")

    x_numeric = np.arange(len(monthly_serious))
    z = np.polyfit(x_numeric, monthly_serious["serious_percent"], 1)
    p = np.poly1d(z)
    ax.plot(monthly_serious["YEAR_MONTH"], p(x_numeric), "darkred", linewidt
            label="Trend line", alpha=0.8)

    ax.set_xlabel("Date", fontsize=12, fontweight="bold")
    ax.set_ylabel("Proportion of Serious Reports (%)", fontsize=12, fontweig
    ax.set_title("COVID-19 VAERS: Monthly Proportion of Serious Outcomes", f
    ax.grid(True, alpha=0.3, linestyle="--")
    ax.legend(fontsize=10)
    ax.set_ylim([0, monthly_serious["serious_percent"].max() * 1.1])

    ax.xaxis.set_major_locator(mdates.MonthLocator(interval=3))
    ax.xaxis.set_major_formatter(mdates.DateFormatter("%Y-%m"))
    plt.xticks(rotation=45, ha="right")

    plt.tight_layout()
    plt.savefig(f"{OUTDIR}/02_monthly_serious_proportion.png", dpi=300, bbox
    print(f"\nSaved: {OUTDIR}/02_monthly_serious_proportion.png")
    plt.show()

    # Save to CSV
    monthly_serious.to_csv(f"{OUTDIR}/monthly_serious_proportion.csv", index
    print(f"Saved: {OUTDIR}/monthly_serious_proportion.csv")

# COMBINED VIEW: COUNT + PROPORTION (DUAL AXIS)

print("\n\n4.3C. COMBINED TIME SERIES (Count + Serious Proportion)")
print("-" * 80)

if "SERIOUS" in df_clean.columns:
    fig, ax1 = plt.subplots(figsize=(14, 7))

    # Left axis: Report count
    color1 = "steelblue"
    ax1.set_xlabel("Date", fontsize=12, fontweight="bold")
    ax1.set_ylabel("Number of Reports", fontsize=12, fontweight="bold", colo
    ax1.plot(monthly_counts["YEAR_MONTH"], monthly_counts["report_count"],
            marker="o", linewidth=2.5, markersize=6, color=color1, label="Re
    ax1.fill_between(monthly_counts["YEAR_MONTH"], monthly_counts["report_co
                    alpha=0.2, color=color1)
    ax1.tick_params(axis="y", labelcolor=color1)
    ax1.grid(True, alpha=0.3, linestyle="--", zorder=1)

    # Right axis: Serious proportion
    ax2 = ax1.twinx()
    color2 = "coral"
    ax2.set_ylabel("Proportion of Serious Reports (%)", fontsize=12, fontwei
    ax2.plot(monthly_serious["YEAR_MONTH"], monthly_serious["serious_percent
            marker="s", linewidth=2.5, markersize=6, color=color2, label="Se
    ax2.fill_between(monthly_serious["YEAR_MONTH"], monthly_serious["serious
```

```
                    alpha=0.2, color=color2)
    ax2.tick_params(axis="y", labelcolor=color2)

    # Title and formatting
    fig.suptitle("COVID-19 VAERS: Monthly Reports and Serious Outcomes",
                fontsize=14, fontweight="bold", y=0.98)

    # Format x-axis
    ax1.xaxis.set_major_locator(mdates.MonthLocator(interval=3))
    ax1.xaxis.set_major_formatter(mdates.DateFormatter("%Y-%m"))
    plt.setp(ax1.xaxis.get_majorticklabels(), rotation=45, ha="right")

    # Legend
    lines1, labels1 = ax1.get_legend_handles_labels()
    lines2, labels2 = ax2.get_legend_handles_labels()
    ax1.legend(lines1 + lines2, labels1 + labels2, loc="upper left", fontsiz

    plt.tight_layout()
    plt.savefig(f"{OUTDIR}/03_combined_count_and_serious.png", dpi=300, bbox
    print(f"\nSaved: {OUTDIR}/03_combined_count_and_serious.png")
    plt.show()

# YEARLY SUMMARY

print("\n\n4.3D. YEARLY SUMMARY STATISTICS")
print("-" * 80)

if "SERIOUS" in df_clean.columns:
    yearly_summary = df_clean.groupby("YEAR").agg({
        "VAERS_ID": "count",
        "SERIOUS": ["sum", "mean"]
    }).reset_index()

    yearly_summary.columns = ["year", "total_reports", "serious_count", "ser
    yearly_summary["serious_percent"] = yearly_summary["serious_rate"] * 100
    yearly_summary = yearly_summary.sort_values("year")

    print("\nYearly breakdown:")
    print(yearly_summary[["year", "total_reports", "serious_count", "serious

    yearly_summary.to_csv(f"{OUTDIR}/yearly_summary.csv", index=False)
    print(f"\nSaved: {OUTDIR}/yearly_summary.csv")
```

```
================================================================================
TIME SERIES ANALYSIS: MONTHLY COVID REPORTS & SERIOUS OUTCOMES
================================================================================

15A. MONTHLY NUMBER OF COVID-19 REPORTS
--------------------------------------------------------------------------------

Date range: 2020-12-01 to 2025-07-01
Total months: 56
Total reports: 986,096

Monthly statistics:
  Mean:     17609 reports/month
  Median:   4396 reports/month
  Min:      262 (in July 2025)
  Max:      101,337 (in April 2021)

✓ Saved: /Users/ariellerothman/Desktop/Capstone/Outputs/01_monthly_report_count.png
```



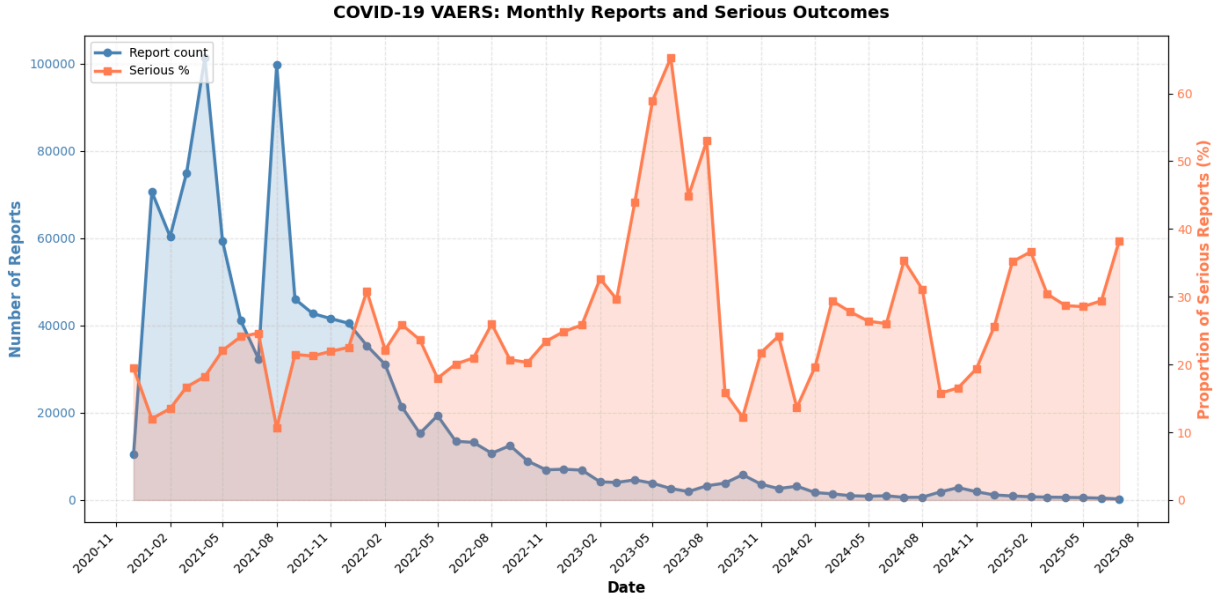

```
✓ Saved: /Users/ariellerothman/Desktop/Capstone/Outputs/monthly_report_count.csv


15B. MONTHLY PROPORTION OF SERIOUS REPORTS
--------------------------------------------------------------------------------

✓ Saved: /Users/ariellerothman/Desktop/Capstone/Outputs/02_monthly_serious_proportion.png
```



✓ Saved: /Users/ariellerothman/Desktop/Capstone/Outputs/monthly_serious_proportion.csv

15C. COMBINED TIME SERIES (Count + Serious Proportion)

✓ Saved: /Users/ariellerothman/Desktop/Capstone/Outputs/03_combined_count_and_serious.png



```
15D. YEARLY SUMMARY STATISTICS
--------------------------------------------------------------------------------
----

Yearly breakdown:
 year  total_reports  serious_count  serious_percent
 2020          10381           2020        19.458626
 2021         710732         126406        17.785326
 2022         195332          46477        23.793848
 2023          47292          15693        33.183202
 2024          18179           3793        20.864734
 2025           4180           1357        32.464115

✓ Saved: /Users/ariellerothman/Desktop/Capstone/Outputs/yearly_summary.csv
```

5. Rule-based comorbidity indicators

This section converts selected free-text fields into structured binary indicators using field-specific keyword dictionaries.

In [ ]:

```
# ============================================================
# 5.1 RULE-BASED COMORBIDITY INDICATORS
# ============================================================
# Purpose: Convert free-text medical narratives into structured binary indic
# Method: Field-specific regex pattern matching with null-token filtering
# Output: Binary columns like HISTORY__cardiovascular, CUR_ILL__diabetes, et


FREE_TEXT_FIELDS = ["HISTORY", "CUR_ILL", "ALLERGIES", "PRIOR_VAX", "LAB_DAT

# Tokens considered "empty" or "no information" for each field
# Prevents false positives from phrases like "no known allergies"
NULL_TOKENS = {
    "none", "no", "n", "na", "n/a", "unknown", "unk", "none known", "not knc
    "nka", "nkda", "no known allergies", "no known drug allergies",
    "denies", "denies allergies", "no allergies", "nil", "negative"
}

# Compiled regex patterns for text cleaning
_keep = re.compile(r"[^a-z0-9\s\-\/]")        # Remove special characters
_ws = re.compile(r"\s+")                       # Collapse whitespace
_url = re.compile(r"http\S+|www\S+|https\S+") # Remove URLs
_email = re.compile(r"\S+@\S+")                # Remove email addresses

def normalize_text(x):
    """
    Normalize free-text medical field to lowercase, clean characters, and re

    Args:
        x (str or NaN): Raw text from medical field.

    Returns:
        str: Cleaned, lowercased text. Empty string if null or contains only
```

```
    Process:
        1. Convert to string and lowercase
        2. Remove URLs and email addresses (common in EMR exports)
        3. Keep only alphanumeric, spaces, hyphens, forward slashes
        4. Collapse multiple spaces
        5. Check final text against null token list

    Example:
        >>> normalize_text("HTN, CAD - no DM")
        'htn cad no dm'

        >>> normalize_text("No known allergies")
        ''  # Returns empty string (recognized as null token)
    """
    if pd.isna(x):
        return ""
    x = str(x).strip().lower()
    if not x:
        return ""
    if x in NULL_TOKENS:
        return ""
    x = _url.sub(" ", x)
    x = _email.sub(" ", x)
    x = _keep.sub(" ", x)
    x = _ws.sub(" ", x).strip()
    if x in NULL_TOKENS:
        return ""
    return x

# Category-specific regex patterns for each text field
# Organized by: {FIELD: {CATEGORY: [patterns]}}
# Each pattern uses word boundaries (\b) to avoid substring matches
CATEGORY_MAP = {
    "CUR_ILL": {
        "acute_infection": [r"\bcovid\b", r"\bsars[-\s]?cov[-\s]?2\b", r"\bi
        "chronic_cardiovascular": [r"\bhypertension\b", r"\bhtn\b", r"\bcad\
        "chronic_respiratory": [r"\basthma\b", r"\bcopd\b", r"\bemphysema\b"
        "metabolic_endocrine": [r"\bdiabetes\b", r"\bdm\b", r"\bdm2\b", r"\b
        "endocrine_thyroid": [r"\bthyroid\b", r"\bhypothyroid\w*\b", r"\bhyp
        "autoimmune_inflammatory": [r"\bautoimmune\b", r"\blupus\b", r"\bsle
        "neurologic": [r"\bseizure\b", r"\bepilepsy\b", r"\bstroke\b", r"\bt
        "neuro_headache": [r"\bmigraine\w*\b", r"\bheadache\w*\b", r"\bclust
        "cancer_immunocompromised": [r"\bcancer\b", r"\bmalignan\w*\b", r"\b
        "lymphatic_edema": [r"\blymph(edema|oedema)\b", r"\bedema\b", r"\bsw
    },
    "HISTORY": {
        "cardiovascular": [r"\bhtn\b", r"\bhypertension\b", r"\bcad\b", r"\b
        "respiratory": [r"\basthma\b", r"\bcopd\b", r"\bemphysema\b", r"\bch
        "metabolic": [r"\bdiabetes\b", r"\bdm\b", r"\bdm2\b", r"\bobesity\b"
        "endocrine_thyroid": [r"\bthyroid\b", r"\bhypothyroid\w*\b", r"\bhyp
        "musculoskeletal": [r"\barthritis\b", r"\bosteoarthritis\b", r"\boa\
        "mental_health": [r"\bdepress\w*\b", r"\banxiety\b", r"\bptsd\b", r"
        "gastrointestinal": [r"\bgerd\b", r"\bacid reflux\b", r"\bgastroesop
        "kidney": [r"\bckd\b", r"\bchronic kidney\b", r"\brenal\b", r"\bdial
        "clotting": [r"\bdvt\b", r"\bpe\b", r"\bthromb\w*\b", r"\bclot\b", r
```

```
        "autoimmune": [r"\bautoimmune\b", r"\blupus\b", r"\bsle\b", r"\bra\b
        "cancer_immunocompromised": [r"\bcancer\b", r"\bmalignan\w*\b", r"\b
        "neurologic": [r"\bstroke\b", r"\btia\b", r"\bseizure\b", r"\bepilep
        "lymphatic_edema": [r"\blymph(edema|oedema)\b", r"\bedema\b"],
    },
    "ALLERGIES": {
        "drug_allergy": [r"\bpenicillin\b", r"\bsulfa\b", r"\bnsaid\b", r"\b
        "food_allergy": [r"\bpeanut\b", r"\bnut\b", r"\begg\b", r"\bmilk\b",
        "latex_other_contact": [r"\blatex\b", r"\bdetergent\b", r"\bfragranc
        "anaphylaxis_history": [r"\banaphylax\w*\b", r"\bepi[-\s]?pen\b", r"
    },
    "PRIOR_VAX": {
        "prior_covid_vax": [r"\bcovid\b", r"\bpfizer\b", r"\bmoderna\b", r"\
        "prior_flu_vax": [r"\bflu\b", r"\binfluenza\b"],
        "prior_other_vax": [r"\bshingrix\b", r"\bshingles\b", r"\btetanus\b"
        "prior_vax_reaction": [r"\breaction\b", r"\ballergic\b", r"\banaphyl
    },
    "LAB_DATA": {
        "cardiac_markers": [r"\btroponin\b", r"\bbnp\b", r"\bck[-\s]?mb\b"],
        "coagulation": [r"\bd[-\s]?dimer\b", r"\binr\b", r"\bptt\b", r"\bfib
        "inflammation": [r"\bcrp\b", r"\besr\b", r"\bferritin\b"],
        "cbc": [r"\bwbc\b", r"\bplatelet\w*\b", r"\bhemoglobin\b", r"\bhgb\b
        "vitals_procedures": [r"\bbp\b", r"\bhr\b", r"\bspo2\b", r"\bultrasc
    },
    "OTHER_MEDS": {
        "anticoagulant": [r"\bwarfarin\b", r"\bheparin\b", r"\beliquis\b", r
        "statin": [r"\bstatin\b", r"\batorvastatin\b", r"\bsimvastatin\b", r
        "immunosuppressant": [r"\bprednisone\b", r"\bsteroid\b", r"\bmethotr
        "diabetes_meds": [r"\binsulin\b", r"\bmetformin\b", r"\bsemaglutide\
        "thyroid_meds": [r"\blevothyroxine\b", r"\bsynthroid\b", r"\bliothyr
        "psych_meds": [r"\bsertraline\b", r"\bfluoxetine\b", r"\bcitalopram\
    },
}

def compile_map(category_map):
    """
    Precompile all regex patterns for efficient matching.

    Args:
        category_map (dict): Nested dict of {field: {category: [pattern_stri

    Returns:
        dict: Same structure with pattern_strings compiled to re.Pattern obj

    Rationale: Compiling patterns once at setup is much faster than compilir
    during each row iteration when used in apply().

    Example:
        >>> compiled = compile_map(CATEGORY_MAP)
        >>> compiled["HISTORY"]["cardiovascular"][0]
        re.Pattern(r'\bhtn\b')
    """
    compiled = {}
    for col, cats in category_map.items():
        compiled[col] = {cat: [re.compile(p) for p in pats] for cat, pats in
    return compiled
```

```
COMPILED_MAP = compile_map(CATEGORY_MAP)

def flag_any(text, patterns):
    """
    Check if any regex pattern matches the text.

    Args:
        text (str): Normalized text to search.
        patterns (list of re.Pattern): Compiled regex patterns.

    Returns:
        int: 1 if any pattern matches, 0 otherwise. Used for binary indicatc

    Example:
        >>> patterns = [re.compile(r'\bdiabetes\b'), re.compile(r'\bdm\b')]
        >>> flag_any("patient has type 2 diabetes", patterns)
        1

        >>> flag_any("patient hypertensive", patterns)
        0
    """
    if not text:
        return 0
    return int(any(p.search(text) for p in patterns))

# Apply categorization to each text field
for col in FREE_TEXT_FIELDS:
    if col not in df_clean.columns:
        continue

    # Normalize the text field
    norm_col = f"{col}_NORM"
    df_clean[norm_col] = df_clean[col].apply(normalize_text)

    # Create missing flag (useful for imputation and model diagnostics)
    df_clean[f"{col}_MISSING"] = (df_clean[norm_col].str.len() == 0).astype(

    # Apply category-specific rules if available
    if col in COMPILED_MAP:
        cat_cols = []
        for cat, pats in COMPILED_MAP[col].items():
            outcol = f"{col}__{cat}"
            df_clean[outcol] = df_clean[norm_col].apply(lambda t: flag_any(t
            cat_cols.append(outcol)

        # Flag records with content that matched NO categories (OTHER bucket
# Useful for quality control and identifying patterns we missed
        df_clean[f"{col}__other"] = (
            (df_clean[f"{col}_MISSING"] == 0) & (df_clean[cat_cols].sum(axis
        ).astype("int8")

# Save engineered comorbidity features ONLY (exclude manufacturer one-hot cc
engineered_cols = [
    c for c in df_clean.columns
    if (("__" in c and not c.startswith("MANU__")) or c.endswith("_MISSING")
```

```
]

comorb_df = df_clean[["VAERS_ID"] + engineered_cols].copy()
comorb_csv_path = os.path.join(OUTDIR, "comorbidity_indicators.csv")
comorb_df.to_csv(comorb_csv_path, index=False)
print("Saved:", comorb_csv_path, "shape:", comorb_df.shape)
```

```
Saved: /Users/ariellerothman/Desktop/Capstone/Outputs/comorbidity_indicator
s.csv shape: (986096, 60)
```

In [ ]:

```
# ============================================================
# 5.2 COMORBIDITY DIAGNOSTICS
# ============================================================

from collections import defaultdict
import pandas as pd

print("=" * 80)
print("SIMPLIFIED COMORBIDITY DIAGNOSTICS")
print("=" * 80)

# A) List all binary indicators by source field

categories = defaultdict(list)

for col in df_clean.columns:
    if "__" in col:
        field, category = col.split("__", 1)
        categories[field].append(category)

print("\nA) INDICATOR CATALOG")
for field in sorted(categories):
    print(f"\n{field}")
    print("-" * len(field))
    for c in sorted(categories[field]):
        print(c)


# B) Why manufacturer indicators show up

manu_cols = [c for c in df_clean.columns if c.startswith("MANU__")]
print("\nB) WHY MANUFACTURER SHOWS UP")
print(
    "Manufacturer appears because the filter '__' includes every double-unde
    "including one-hot manufacturer columns like MANU__PFIZER/BIONTECH."
)
print(f"Manufacturer indicator columns detected: {len(manu_cols)}")
if manu_cols:
    print("Examples:", manu_cols[:10])


# C) Explore indicator groups (example: HISTORY__kidney)

print("\nC) SAMPLE RECORDS FOR ONE INDICATOR")
example_indicator = "HISTORY__kidney"
if example_indicator in df_clean.columns and "HISTORY" in df_clean.columns:
```

```
    n = min(20, int((df_clean[example_indicator] == 1).sum()))
    if n > 0:
        display(df_clean.loc[df_clean[example_indicator] == 1, ["HISTORY"]].
    else:
        print(f"No positive rows found for {example_indicator}.")
else:
    print(f"Column missing: {example_indicator} or HISTORY")


# D) Summary table for indicator prevalence
print("\nD) INDICATOR PREVALENCE SUMMARY")
indicator_cols = [
    c for c in df_clean.columns
    if "__" in c and not c.startswith("MANU__")
]

summary = (
    df_clean[indicator_cols]
    .sum()
    .sort_values(ascending=False)
    .rename("count")
    .to_frame()
)

summary["percent"] = summary["count"] / len(df_clean) * 100

display(summary)

# Optional export
summary.to_csv(f"{OUTDIR}/simplified_indicator_summary.csv")
print(f"Saved: {OUTDIR}/simplified_indicator_summary.csv")

# ---------------------------------------------------------------------
# E) Show examples for each FIELD__other bucket
# ---------------------------------------------------------------------
print("\nE) FIELD__other EXAMPLES")

fields = ["HISTORY", "CUR_ILL", "ALLERGIES", "PRIOR_VAX", "LAB_DATA", "OTHER

for field in fields:
    other_col = f"{field}__other"

    if other_col not in df_clean.columns or field not in df_clean.columns:
        continue

    candidates = df_clean.loc[df_clean[other_col] == 1, field].dropna()

    print("\n", field, "OTHER examples")

    if len(candidates) == 0:
        print("(none)")
        continue

    n = min(10, len(candidates))
    print(candidates.sample(n, random_state=42).to_string(index=False))
```

==========
SIMPLIFIED COMORBIDITY DIAGNOSTICS
==========

A) INDICATOR CATALOG

ALLERGIES
---
anaphylaxis_history
drug_allergy
food_allergy
latex_other_contact
other

CUR_ILL
---
acute_infection
autoimmune_inflammatory
cancer_immunocompromised
chronic_cardiovascular
chronic_respiratory
endocrine_thyroid
lymphatic_edema
metabolic_endocrine
neuro_headache
neurologic
other

HISTORY
---
autoimmune
cancer_immunocompromised
cardiovascular
clotting
endocrine_thyroid
gastrointestinal
kidney
lymphatic_edema
mental_health
metabolic
musculoskeletal
neurologic
other
respiratory

LAB_DATA
---
cardiac_markers
cbc
coagulation
inflammation
other
vitals_procedures

```
MANU
----
JANSSEN
MODERNA
NOVAVAX
PFIZER_BIONTECH
UNKNOWN_MANUFACTURER

OTHER_MEDS
----------
anticoagulant
diabetes_meds
immunosuppressant
other
psych_meds
statin
thyroid_meds

PRIOR_VAX
---------
other
prior_covid_vax
prior_flu_vax
prior_other_vax
prior_vax_reaction

B) WHY MANUFACTURER SHOWS UP
Manufacturer appears because the filter '__' includes every double-underscor
e binary column, including one-hot manufacturer columns like MANU__PFIZER/BI
ONTECH.
Manufacturer indicator columns detected: 5
Examples: ['MANU__JANSSEN', 'MANU__MODERNA', 'MANU__NOVAVAX', 'MANU__PFIZER_
BIONTECH', 'MANU__UNKNOWN_MANUFACTURER']

C) SAMPLE RECORDS FOR ONE INDICATOR
```

	HISTORY
919812	Asthma, UTIs, Medullary cystic kidney disease,...
32019	Advanced dementia with severe violent behavior...
477431	Gastroesophageal reflux disease 03/16/2021 Unk...
593200	Acidic stomach, Stage 3 CKD
407837	DIABETES MELLITUS, TYPE 2. CHRONIC KIDNEY DISE...
912348	Cardiovascular Hypertension Paroxysmal atrial ...
607125	Acid reflux, Arthritis, Eczema, Enlarged prost...
571557	Diabetes, Cardiovascular disease, Renal disease
163048	Allergy induced asthma. Renal failure than ha...
321342	lung disease, heart disease, renal disease
813112	lung transplant recipient, CAD< hypertension, ...
596453	Nervous Cerebrovascular accident (CVA) due to ...
650119	Systemic Lupus Erythematous (lung and kidney) ...
394854	Diabetes mellitus, heart disease, immune suppr...
567565	Medical History/Concurrent Conditions: Renal d...
685510	Benign Prostatic Hyperplasia With Lower Urinar...
851136	End stage renal disease, type 2 diabetes melli...
58728	Atrial fibrillation, high blood pressure, high...
795670	OSA DM type 2 peripheral dema OA obesity insom...
628676	Medical History/Concurrent Conditions: Kidney ...

D) INDICATOR PREVALENCE SUMMARY

	count	percent
OTHER_MEDS__other	271700	27.553098
LAB_DATA__other	195633	19.839143
HISTORY__other	172946	17.538455
ALLERGIES__other	143783	14.581035
ALLERGIES__drug_allergy	77583	7.867692
CUR_ILL__other	71218	7.222218
HISTORY__cardiovascular	63285	6.417732
HISTORY__respiratory	59886	6.073040
HISTORY__metabolic	56111	5.690217
LAB_DATA__vitals_procedures	53601	5.435678
OTHER_MEDS__statin	43514	4.412755
OTHER_MEDS__thyroid_meds	41952	4.254353
HISTORY__endocrine_thyroid	37831	3.836442
HISTORY__musculoskeletal	36222	3.673273
HISTORY__mental_health	35486	3.598635
OTHER_MEDS__psych_meds	29587	3.000418
HISTORY__neurologic	27809	2.820111
HISTORY__gastrointestinal	25701	2.606339
OTHER_MEDS__diabetes_meds	24974	2.532613
HISTORY__cancer_immunocompromised	19035	1.930339
CUR_ILL__acute_infection	15592	1.581185
ALLERGIES__latex_other_contact	14381	1.458377
PRIOR_VAX__prior_covid_vax	13845	1.404022
PRIOR_VAX__other	13457	1.364674
ALLERGIES__food_allergy	12825	1.300583
PRIOR_VAX__prior_flu_vax	12237	1.240954
OTHER_MEDS__anticoagulant	11166	1.132344
HISTORY__autoimmune	9968	1.010855
CUR_ILL__chronic_cardiovascular	9576	0.971102
HISTORY__kidney	9310	0.944127
CUR_ILL__metabolic_endocrine	9293	0.942403
CUR_ILL__chronic_respiratory	8725	0.884802

	count	percent
OTHER_MEDS__immunosuppressant	7235	0.733701
LAB_DATA__cbc	6903	0.700033
PRIOR_VAX__prior_other_vax	6362	0.645170
LAB_DATA__cardiac_markers	6337	0.642635
CUR_ILL__autoimmune_inflammatory	5644	0.572358
LAB_DATA__coagulation	5415	0.549135
CUR_ILL__endocrine_thyroid	5406	0.548222
HISTORY__clotting	4978	0.504819
CUR_ILL__cancer_immunocompromised	4827	0.489506
LAB_DATA__inflammation	4523	0.458677
CUR_ILL__neuro_headache	4080	0.413753
PRIOR_VAX__prior_vax_reaction	3945	0.400062
ALLERGIES__anaphylaxis_history	3343	0.339014
CUR_ILL__neurologic	3099	0.314270
HISTORY__lymphatic_edema	1876	0.190245
CUR_ILL__lymphatic_edema	834	0.084576

```
Saved: /Users/ariellerothman/Desktop/Capstone/Outputs/simplified_indicator_s
ummary.csv

E) FIELD__other EXAMPLES

 HISTORY OTHER examples
                            non hodgkins lymphonia
chronic gastritis, tachycardia, kidney stones. ...
Comments: List of non-encoded Patient Relevant ...
                                recurring shingles
                                      BMI elevated
Patient has idiopathic hives sometimes and also...
  Medical History/Concurrent Conditions: COVID-19.
Medical History/Concurrent Conditions: COVID-19...
Comments: No medical history was provided by th...
                      Barrett's esophagus syndrome

 CUR_ILL OTHER examples
                 period started day of vaccination
                      Anxiety; Blood pressure high
            Alcohol use (weekly maybe); Non-smoker
Bee sting hypersensitivity; Drug allergy (torad...
Allergy to vaccine; Drug allergy (NSAIDS); Drug...
         Sulfonamide allergy (Allergy sulfa drugs)
                                     None reported
                                         Menopause
Was not sick at time of vaccination. However I ...
                                      not reported

 ALLERGIES OTHER examples
                             Demerol, statins, PCN
Dust, mold, animals, and some foods, outdoor al...
Positive for ?variant of unknown significance? ...
                                           Codeine
                                           Demerol
                                         dust mites
Cipro, Gluten, ASAIDs,Squash, Cassein in cheese...
                        Erythromycin,  amoxicillin
                                    Pcn and batrim
                              Codeine Erythromycin

 PRIOR_VAX OTHER examples
                                         dizziness
                                  Sore Arm, Fatigue
Anthrax, Smallpox, Malaria, really weird lucid ...
                                         Pneumonia
                                  Just mild nausea
                                   Anthrax in 2009
                            Gardasil, Anaphylactic
Johnson and Johnson vaccine received previous a...
                       Fever, chills, and headache
yes, states she has an autoimmune disease and t...

 LAB_DATA OTHER examples
                                       none needed
                                      Kidney Scans
```

```
            MEDIC STATED HE HAD LOW BLOOD PRESSURE
Will consult with physician (see above) re:test...
                                                 0
Test Date: 20210318; Test Name: Pulse rate; Res...
Test Name: COVID-19 TEST; Test Result: Positive...
14JUL2022 - COVID-19 At Home Test - Positive; 0...
                11/07/21 SARS-CoV-2 by NAA detected
Test Date: 20210115; Test Name: Blood glucose; ...

 OTHER_MEDS OTHER examples
CLARITIN 12 HOUR 10MG (taken at @2:00pm BABY AS...
                                   No active meds
                                        CARVEDIOL
                               Multivitamin and D3
                                      Multivitamin
                               Daily multi vitamin
                     Birth Control (Yaz), Benadryl
Tylenol took one dose of 7.5 ml in the morning ...
           propranolol 80 mg ER and Losartan 50 mg
           metoprolol succinate 50mg, viibryd 40mg
```

In [ ]:

```
# ============================================================
# 5.3 IMPUTE MISSING VALUES FOR MODEL-READY EXPORT
# ============================================================

df_clean_imputed = df_clean.copy()

# Snapshot missingness before imputation
missing_before = int(df_clean_imputed.isna().sum().sum())
print(f"Total missing values BEFORE imputation: {missing_before:,}")

# 0) Enforce numeric dtypes for known numeric model fields
# This prevents string artifacts (e.g., "UNKNOWN") from polluting numeric co
expected_numeric_cols = [
    "AGE_YRS", "HOSPDAYS", "NUMDAYS", "ONSET_INTERVAL",
    "MAX_DOSE", "DOSE_COUNT", "MULTI_DOSE", "UNKNOWN_DOSE", "MULTI_MANUFACTU
    "YEAR", "MONTH", "SERIOUS", "SEVERE"
]
for col in expected_numeric_cols:
    if col in df_clean_imputed.columns:
        df_clean_imputed[col] = pd.to_numeric(df_clean_imputed[col], errors=

# 1) Binary indicators: fill with 0
binary_indicator_cols = [
    c for c in df_clean_imputed.columns
    if ("__" in c) or c.endswith("_MISSING")
]
for col in binary_indicator_cols:
    if col in df_clean_imputed.columns:
        df_clean_imputed[col] = pd.to_numeric(df_clean_imputed[col], errors=

# 2) Numeric features: median imputation (except VAERS_ID)
numeric_cols = df_clean_imputed.select_dtypes(include=["number"]).columns.to
numeric_fill_cols = [c for c in numeric_cols if c not in binary_indicator_co
for col in numeric_fill_cols:
    if df_clean_imputed[col].isna().any():
```

```
        med = df_clean_imputed[col].median()
        if pd.isna(med):
            med = 0
        df_clean_imputed[col] = df_clean_imputed[col].fillna(med)

# 3) Free-text columns: empty string imputation
text_cols = [
    "SYMPTOM_TEXT", "LAB_DATA", "OTHER_MEDS", "CUR_ILL", "HISTORY", "PRIOR_V
    "HISTORY_NORM", "CUR_ILL_NORM", "ALLERGIES_NORM", "PRIOR_VAX_NORM", "LAB
]
for col in text_cols:
    if col in df_clean_imputed.columns:
        df_clean_imputed[col] = df_clean_imputed[col].fillna("")

# 4) Remaining object/string columns: 'UNKNOWN'
# (Only non-numeric-like columns should reach this step.)
obj_cols = df_clean_imputed.select_dtypes(include=["object", "string"]).colu
for col in obj_cols:
    if col in text_cols:
        continue
    if df_clean_imputed[col].isna().any():
        df_clean_imputed[col] = df_clean_imputed[col].fillna("UNKNOWN")

missing_after = int(df_clean_imputed.isna().sum().sum())
print(f"Total missing values AFTER imputation:  {missing_after:,}")
print(f"Missing values reduced by:             {missing_before - missing_aft
print(f"Imputed dataframe shape: {df_clean_imputed.shape}")

# Quick audit for remaining missingness
remaining_missing = (
    df_clean_imputed.isna().sum()
    .loc[lambda s: s > 0]
    .sort_values(ascending=False)
)
if len(remaining_missing):
    print("\nRemaining missing values by column:")
    print(remaining_missing)
else:
    print("\nNo remaining missing values.")
```

```
Total missing values BEFORE imputation: 6,750,567
Total missing values AFTER imputation:  1,140,074
Missing values reduced by:             5,610,493
Imputed dataframe shape: (986096, 113)

Remaining missing values by column:
DATEDIED      968921
ONSET_DATE     97200
VAX_DATE       73953
dtype: int64
```

In [ ]:

```
# ============================================================
# 5.4 SAVE FINAL DATASETS (ENGINEERED + IMPUTED)
# ============================================================

# Save engineered dataset (pre-imputation snapshot)
```

```python
engineered_csv_path = os.path.join(OUTDIR, "df_clean_engineered.csv")
df_clean.to_csv(engineered_csv_path, index=False)
print("Saved engineered dataset:", engineered_csv_path)
print("Engineered shape:", df_clean.shape)

# Save imputed dataset for downstream modeling notebooks
imputed_csv_path = os.path.join(OUTDIR, "df_clean_imputed.csv")
df_clean_imputed.to_csv(imputed_csv_path, index=False)
print("Saved imputed dataset:", imputed_csv_path)
print("Imputed shape:", df_clean_imputed.shape)

# Data dictionary for imputed export
data_dict = pd.DataFrame({
    "column": df_clean_imputed.columns,
    "dtype": df_clean_imputed.dtypes.astype(str),
    "non_null_count": df_clean_imputed.count(),
    "null_count": df_clean_imputed.isna().sum()
})
dict_path = os.path.join(OUTDIR, "data_dictionary.csv")
data_dict.to_csv(dict_path, index=False)
print("\nSaved data dictionary:", dict_path)
```

```
Saved engineered dataset: /Users/ariellerothman/Desktop/Capstone/Outputs/df_
clean_engineered.csv
Engineered shape: (986096, 113)
Saved imputed dataset: /Users/ariellerothman/Desktop/Capstone/Outputs/df_cle
an_imputed.csv
Imputed shape: (986096, 113)

Saved data dictionary: /Users/ariellerothman/Desktop/Capstone/Outputs/data_d
ictionary.csv
```